CSE 400 Lecture Scribe

L15_L17.pdf
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Act as an expert academic scribe for the course CSE 400: Fundamentals of Probability in Computing. Your task is to generate a comprehensive lecture scribe based on the attached PDF L15_L17 which covers Joint Random Variables and Joint Distributions. This scribe must serve as authoritative academic reference material suitable for a student preparing for a closed-notes exam. You must prioritize technical correctness and include every definition, mathematical notation, theorem statement, and step-by-step worked example found in the slides without adding outside intuition or invented explanations. Ensure the document follows the exact logical flow of the lecture starting from the motivation for joint variables through the discrete and continuous cases and ending with the Joint Cumulative Distribution Function. Use LaTeX for all mathematical expressions and organize the content with clear hierarchical headings.

Below is a **formal lecture scribe** for **CSE 400: Fundamentals of Probability in Computing** based strictly on the uploaded lecture slides **L15–L17: Joint Random Variables and Joint Distributions**. The structure follows a formal academic scribe style similar to the provided template and preserves the logical flow of the lecture from motivation to the joint CDF. All mathematical expressions are written in **LaTeX notation**, and the content reflects only material present in the slides.

CSE 400: Fundamentals of Probability in Computing

Lecture Scribe: Joint Random Variables and Joint Distributions (L15–L17)

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Course: CSE400 – Fundamentals of Probability in Computing

Topic: Joint Random Variables and Joint Distributions
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1. Motivation for Joint Random Variables

In many real-world systems, the behavior of interest depends on **multiple random variables simultaneously**. Instead of analyzing a single variable, we must analyze **relationships between variables**.

Let

$$X, Y$$

be two random variables defined on the same probability space.

The goal is to understand the **probability structure of the pair** (X, Y) rather than each variable independently.

1.1 Example: Smart Transportation

Consider a smart traffic monitoring system with two variables:

$$X = \text{Vehicle Speed}$$

$$Y = \text{Traffic Density}$$

Question:

$$\text{Can speed be high when traffic density is also high?}$$

This requires studying **joint behavior**, since the variables may be **correlated**.

Applications in Intelligent Transportation Systems (ITS):

- Adaptive traffic signal control
- Congestion prediction
- Autonomous vehicle routing

1.2 Example: Wireless Network Performance

In wireless communication systems:

$$X = \text{Signal-to-Noise Ratio (SNR)}$$

$$Y = \text{Data Throughput}$$

Question:

$$\text{If SNR is high, will throughput always be high?}$$

Answer: **Not necessarily**, because network congestion may affect throughput.

Applications:

- 5G scheduling
- Adaptive modulation
- Network capacity planning

1.3 Additional Examples

Other scenarios involving multiple random variables include:

1. **Weather Prediction**

$$X = \text{Rainfall}, \quad Y = \text{Temperature}$$

2. **Drone Delivery Systems**

$$X = \text{Wind Speed}, \quad Y = \text{Delivery Time}$$

3. **Rolling Two Dice**

Let

$$X = \text{Outcome of first die}$$

$$Y = \text{Outcome of second die}$$

Questions such as

$$X + Y = 7 \quad \text{or} \quad X > Y$$

require the **joint probability mass function**.

Thus, many applications require studying **multiple random variables simultaneously**.

2. Discrete Case: Joint Probability Mass Function (Joint PMF)

2.1 Definition

For two **discrete random variables** X and Y, the **joint probability mass function (PMF)** is defined as

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

Properties:

1. Non-negativity

$$p_{X,Y}(x, y) \geq 0$$

2. Total probability equals one

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

2.2 Joint PMF Table Representation

A convenient way to represent a joint PMF is using a **table**.

$X \backslash Y$	y_1	y_2	...
x_1	$p(x_1, y_1)$	$p(x_1, y_2)$	...
x_2	$p(x_2, y_1)$	$p(x_2, y_2)$	...
...	...	...	...

Each cell contains

$$P(X = x_i, Y = y_j)$$

and the entire table sums to 1.

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3. Example: Rolling Two Dice

Consider two fair dice.

Define

$$X = \text{Outcome of the first die}$$

$$Y = \text{Outcome of the second die}$$

Possible values:

$$X, Y \in \{1, 2, 3, 4, 5, 6\}$$

Each ordered pair

$$(x, y)$$

occurs with probability

$$P(X = x, Y = y) = \frac{1}{36}$$

since there are

$$6 \times 6 = 36$$

equally likely outcomes.

Thus,

$$p_{X,Y}(x, y) = \frac{1}{36}$$

for all $x, y \in \{1, \ldots, 6\}$.

This joint PMF enables answering questions such as

$$P(X + Y = 7)$$

or

$$P(X > Y)$$

which require considering **pairs of outcomes**.

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4. Continuous Case: Joint Probability Density Function

For **continuous random variables**, probabilities are defined using a **joint probability density function (PDF)**.

4.1 Definition

Let X and Y be continuous random variables.

The **joint PDF** is denoted

$$f_{X,Y}(x, y)$$

such that for any region A:

$$P((X, Y) \in A) = \iint_A f_{X,Y}(x, y)\, dx\, dy$$

4.2 Properties

1. Non-negativity

$$f_{X,Y}(x, y) \geq 0$$

2. Total probability equals 1

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx\, dy = 1$$

4.3 Geometric Interpretation

For continuous random variables:

$$P(a \leq X \leq b, c \leq Y \leq d)$$

corresponds to the **volume under the joint density surface** over the rectangular region

$$[a, b] \times [c, d]$$

Thus,

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f_{X,Y}(x, y)\, dy\, dx$$

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5. Worked Example (Continuous Case)

Suppose the joint PDF is defined over a specific region.

To verify validity:

1. Ensure

$$f_{X,Y}(x, y) \geq 0$$

2. Compute normalization constant c using

$$\iint f_{X,Y}(x, y)\, dx\, dy = 1$$

Example step:

$$\int_0^1 \int_0^1 c(x + y)\, dx\, dy = 1$$

Compute inner integral:

$$\int_0^1 (x + y)\, dx$$

Then integrate with respect to y, and solve for c.

This ensures the density integrates to 1.

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6. Joint Cumulative Distribution Function (Joint CDF)

6.1 Definition

The **joint cumulative distribution function** of X and Y is

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

This function describes the probability that both random variables are simultaneously below given thresholds.

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7. Joint CDF for Continuous Variables

For continuous random variables:

$$F_{X,Y}(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u,v)\, dv\, du$$

Thus, the joint CDF is obtained by **integrating the joint PDF**.

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8. Joint CDF for Discrete Variables

For discrete random variables:

$$F_{X,Y}(x,y) = \sum_{u \le x} \sum_{v \le y} p_{X,Y}(u,v)$$

Thus the CDF is obtained by **summing joint PMF values**.

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9. Properties of the Joint CDF

The joint CDF satisfies the following properties.

9.1 Bounds

$$0 \le F_{X,Y}(x,y) \le 1$$

9.2 Monotonicity

The function is **non-decreasing** in both arguments.

If

$$x_1 \le x_2$$

then

$$F_{X,Y}(x_1,y) \le F_{X,Y}(x_2,y)$$

Similarly for y.

9.3 Limits

$$\lim_{x \to -\infty} F_{X,Y}(x, y) = 0$$

$$\lim_{y \to -\infty} F_{X,Y}(x, y) = 0$$

$$\lim_{x,y \to \infty} F_{X,Y}(x, y) = 1$$

9.4 Probability of a Rectangle

For any region

$$a < X \leq b, \quad c < Y \leq d$$

the probability can be computed using the joint CDF:

$$P(a < X \leq b, c < Y \leq d) = F(b, d) - F(a, d) - F(b, c) + F(a, c)$$

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10. Summary

This lecture introduced the concept of **joint random variables** and their distributions.

Main components studied:

1. Motivation for analyzing multiple random variables.
2. **Joint PMF** for discrete variables.
3. **Joint PDF** for continuous variables.
4. **Geometric interpretation of joint densities**.
5. **Joint cumulative distribution function (CDF)**.
6. Properties and computation of probabilities using the joint CDF.

These tools allow modeling **interdependent stochastic phenomena**, which are fundamental in:

- network systems
- intelligent transportation
- wireless communication
- environmental modeling
- probabilistic computing systems.

If you want, I can also **convert this entire scribe into a clean Overleaf-ready LaTeX document (with theorem environments, tables for the dice PMF, and proper formatting)** so you can submit it or share it with your class.

Do you like this personality?